Final Exam

1. Order the following market structures from **most competitive (1) to least competitive (4)**. /0.5

   ___ 1. Perfect Competition

   ___ 2. Oligopoly

   ___ 3. Monopoly

   ___ 4. Monopolistic Competition

2. Select all the conditions for a "**perfectly competitive**" market structure. /0.5

   Select all correct answers

   - ☐ No barriers to entry or transaction costs (free entry and exit).
   - ☐ No individual seller is able to effect market price on their own.
   - ☐ Buyers and Sellers have all needed information.
   - ☐ There are high quality products made from 1 or 2 providers.
   - ☐ Businesses can make the most possible.
   - ☐ The government decides who gets what.
   - ☐ Many buyers and sellers.
   - ☐ All products are identical in that market and all products have equal value (Interchangeable).

3. A small organic farmer operates in a market structure where the prevailing equilibrium price is $5.47 per bushel for their corn. /0.5

If the farmer decides to sell their identical corn crop for $6.00 per bushel, what is the most likely outcome?

- ○ The farmer will successfully increase their economic profit margin on every bushel sold.
- ○ The farmer will sell zero bushels because buyers can purchase identical corn for $5.47 from competitors
- ○ The farmer will capture all buyers currently buying corn at prices below $6.00
- ○ The farmer's total revenue will increase due to the higher price, despite selling a lower quantity
- ○ The farmer will still sell their usual quantity because the small change in price does not affect overall market supply or demand.

4. Given the data provided in the table below, what will the marginal revenue equal for production at quantity (Q) level 4? /0.5

Q	P	TC	TR	MR	MC	Profit
0	$5	$9				
1	$5	$10				
2	$5	$12				
3	$5	$15				
4	$5	$19				
5	$5	$24				
6	$5	$30				
7	$5	$45				

- ○ $20
- ○ $1
- ○ $5
- ○ $15

5. An individual competitive firm, in determining its optimal output quantity, must consider which of the following? /0.5

Select all correct answers

- ☐ The price elasticity of demand for the entire industry
- ☐ The market price for its product, which determines its marginal revenue
- ☐ The prices of inputs (labor, raw materials) used in production, which determine cost
- ☐ Whether the market price is sufficient to cover average variable costs in the short run
- ☐ The quantity where Marginal Revenue equals Marginal Cost (MR = MC)

6. Match the Terms with the correct Economic Concept. /0.5

___ 1. Small Relative to Market	A. Firms have full and transparent knowledge regarding the goods being exchanged.
___ 2. Perfectly Elastic Demand	B. A business can increase or decrease its output without influencing the overall market price.
___ 3. Perfect Information	C. The demand curve faced by the individual firm is a horizontal line.
___ 4. Identical Products	D. The output produced by one firm is highly interchangeable with the output of any other firm
___ 5. Price Taker	E. The market pressure forces the firm to accept the equilibrium price

7. A perfectly competitive firm is currently producing a quantity where Marginal Cost (MC) is $15 and Marginal Revenue (MR) is $20. /0.5

If Average Total Cost (ATC) is $18, what action should the firm take to maximize profits?

- ◯ Decrease output, because ATC is currently higher than MR.
- ◯ Decrease output until MR = ATC
- ◯ Increase output, because the additional revenue gained exceeds the additional cost
- ◯ Shut down immediately, as the firm is making a loss.
- ◯ Maintain the current output level, as it is earning positive economic profits (P > ATC

8. In the long run, perfectly competitive firms achieve **productive efficiency**. /0.5

This condition is met because:

- ◯ The zero economic profit condition means the firms' total revenue exactly equals their variable costs.
- ◯ Firms maximize utility rather than profit.
- ◯ Firms produce and sell goods at the highest possible marginal cost.
- ◯ The competitive process of entry and exit drives the price down to the minimum of the long-run average cost curve
- ◯ P is set equal to MR, maximizing total revenue.

9. A company that specializes in high-end specialty chemicals maintains its dominant market position by controlling access to the only known concentrated deposit of a rare mineral input necessary for its product. /0.5

Additionally, the company invests heavily in aggressive, targeted marketing campaigns that make it prohibitively expensive for new firms to gain brand recognition.

Which two types of barriers to entry are primarily protecting this firm?

- ◯ Trade Secret and Exclusive Licensing
- ◯ Natural Monopoly and Legal Monopoly (Copyright)
- ◯ Control of a Physical Resource and Intimidating Potential Competitors (Advertising)
- ◯ Natural Monopoly and Predatory Pricing
- ◯ Legal Monopoly (Patent) and Economies of Scale

10. Which of the following legal instruments are specifically designed to **promote innovation** by granting limited monopoly power over intellectual property? /0.5

Select all correct answers

- ☐ Trademarks
- ☐ Copyrights
- ☐ Patents
- ☐ Trade Secrets
- ☐ Deregulation policies

11. Match the correct Barrier To Entry to each Scenario. /0.5

___ 1. Legal Monopoly (Patent)

___ 2. Control of a Physical Resource

___ 3. Intimidation (Predatory Pricing)

___ 4. Natural Monopoly

A. Regional water company (duplicative infrastructure costly).

B. Corporation controls most global mining of a scarce gemstone.

C. Government grants 20 years exclusive right to sell new drug.

D. Large retailer sells below cost after competitor enters.

12. To calculate and ensure they are maximizing total economic profit, a monopolist must fulfill which of the following conditions or steps? /0.5

Select all correct answers

- ☐ Determine the quantity of output where Marginal Revenue equals Marginal Cost (MR = MC).
- ☐ Identify the price corresponding to the quantity chosen in step A by referencing the Average Cost curve.
- ☐ Produce where Price (P) equals Marginal Cost (MC) to achieve allocative efficiency.
- ☐ Ensure that price is above the average cost at the profit-maximizing output level.
- ☐ Calculate marginal profit (MR - MC) and confirm it equals zero at the chosen output level.

13. Match the Economic Concept to its Significance in Monopoly Decision Making. /0.5

___ 1. Price is greater than Marginal Cost (P > MC) at optimal output.

___ 2. The point on the Demand Curve directly above the MR = MC intersection.

___ 3. The calculation of Total Revenue minus Total Cost.

___ 4. The intersection of Marginal Revenue and Marginal Cost curves (MR = MC).

A. Determines the profit-maximizing quantity of output.

B. Determines the highest price consumers are willing to pay for the profit-maximizing quantity.

C. Represents the firm's overall highest achievable profit level.

D. A condition demonstrating the lack of allocative efficiency.

14. Which of the following barriers to entry, when significant, enable a firm currently earning above-normal economic profits to maintain those profits in the long run? /0.5

Select all correct answers

- ☐ The high but surmountable cost of renting retail space (a simple fixed cost)
- ☐ High economies of scale relative to market demand, leading to a natural monopoly.
- ☐ The firm having full control over a scarce and necessary mineral input
- ☐ Government-granted copyrights protecting the original authorship of a software code.
- ☐ A successful, multi-decade history of maintaining a famous trade secret, such as a proprietary formula.

15. Product differentiation in monopolistic competition is closely related to the degree of variety available in the economy. /0.5

Which of the following demonstrate differentiation through means other than the physical good itself?

Select all correct answers

- ☐ A grocery store that is the only one located within a densely populated urban area.
- ☐ A dry cleaner that offers a money-back guarantee of satisfaction.
- ☐ Using advanced manufacturing processes to ensure the product has a non-stick surface.
- ☐ Extensive advertising that encourages consumers to associate the brand with luxury and status.
- ☐ A non-shrink, newly redesigned fabric used in athletic wear.

16. If a monopolistically competitive firm is making positive economic profits in the short run, the process of long-run adjustment ensures that: /0.5

- ○ New competitors enter, shifting the existing firm's perceived demand curve to the left until P = AC.
- ○ Existing firms raise their prices to maximize monopoly profits, attracting further entry
- ○ Barriers to entry increase, allowing the original firm to sustain positive economic profits.
- ○ The firm's marginal cost curve shifts upward due to increased advertising costs.
- ○ The firm increases output until price equals marginal cost (P = MC).

17. In the framework of monopolistic competition, what is the fundamental goal of advertising? /0.5

- ○ To ensure the firm operates solely on the basis of lowest possible average cost.
- ○ To convince potential buyers that the firm's product is meaningfully different from competitors' offerings.
- ○ To achieve perfect elasticity in the perceived demand curve.
- ○ To increase total costs until they equal total revenues.
- ○ To guarantee that the firm achieves productive efficiency.

18. A monopolistically competitive firm operating in the short run finds that at its current output level, Marginal Revenue ($20) is greater than Marginal Cost ($15), and Price ($30) is greater than Average Total Cost ($25). /0.5

To maximize profit, the firm should:

- ◯ Decrease output and raise price, since the current price is above average total cost.
- ◯ Increase output and keep the current price, since MR is greater than MC.
- ◯ Decrease output until MR = MC, resulting in zero economic profit.
- ◯ Maintain the current output and price, as the firm is already making a profit (P > ATC).
- ◯ Increase output and find the new price on the demand curve corresponding to the new MR=MC quantity.

19. Match the characteristic to the correct market structure. /0.5

___ 1. Oligopoly

___ 2. Monopolistic Competition

___ 3. Monopoly

___ 4. Product Differentiation

___ 5. Economies of Scale

A. Defined by a large number of firms and differentiated products.

B. Defined by a single firm selling a product with no close substitutes.

C. Defined by a small number of large firms and mutual interdependence.

D. A barrier to entry that ensures only large firms can operate at low average cost.

E. A feature that can be combined with cost advantages to create a brand advantage.

20. A key benefit that monopolistic competition offers to society, which is generally absent in perfect competition, is: /0.5

- ◯ The complete elimination of advertising costs.
- ◯ Guaranteed long-run productive efficiency.
- ◯ Extremely high barriers to entry protecting firms from competition.
- ◯ Guaranteed allocative efficiency (P = MC).
- ◯ A desired degree of product variety, innovation, and differentiated shopping experiences.

21. When evaluating the tradeoffs of monopolistic competition, the downward-sloping demand curve leads directly to which two efficiency shortcomings? /0.5

- ◯ Zero economic profit and high barriers to entry.
- ◯ High innovation rates and excessive variety.
- ◯ Perfect elasticity and low marketing expenditure.
- ◯ Productive inefficiency (not min AC) and allocative inefficiency (P > MC).
- ◯ Mutual interdependence and the risk of collusion.

22. The game matrix below shows the payoffs associated with the strategies of two players. Assume that the objective of each player is to earn the largest payoff. /0.5

		Joseph	
		go left	go right
Asil	go left	\$4 , \$5	\$7 , \$9
	go right	\$10 , \$6	\$8 , \$3

What is Asil's payoff if she goes right when Joseph goes left?

- ◯ \$10
- ◯ \$6
- ◯ \$9
- ◯ \$3

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23. Two rival firms in a duopoly, Alpha and Beta, face a decision regarding production levels: High Output or Low Output. Their payoff matrix (profits in \$ thousands) is shown below: /0.5

	Beta: Low Output	**Beta: High Output**
Alpha: Low Output	A: \$1000, B: \$1000	A: \$200, B: \$1500
Alpha: High Output	A: \$1500, B: \$200	A: \$400, B: \$400

If both firms pursue their individual self-interest independently, what is the resulting outcome?

- ◯ Alpha earns \$1,500 and Beta earns \$200.
- ◯ Alpha chooses Low Output and Beta chooses High Output.
- ◯ Both firms achieve the maximum combined profit of \$2,000.
- ◯ Both firms choose Low Output, resulting in \$1,000 profit each.
- ◯ Both firms choose High Output, resulting in \$400 profit each.

24. A group of U.S. firms that secretly coordinate to reduce industry output and keep prices high, without signing a formal agreement, is engaging in: /0.5

- ◯ Collusion, which is illegal anti-competitive behavior.
- ◯ Price taking behavior.
- ◯ Allocative efficiency.
- ◯ Perfect competition.
- ◯ Operating under a strictly enforced cartel structure.

25. Oligopolistic markets are characterized by: /0.5

Select all correct answers

- ☐ A small number of large firms dominating sales.
- ☐ Mutual interdependence among firms' output, price, and advertising decisions.
- ☐ Price taking behavior similar to perfect competition
- ☐ A large number of small firms.
- ☐ High barriers to entry.

26. The prisoner's dilemma scenario is an effective model for oligopoly behavior because: /0.5

Select all correct answers

- ☐ Oligopolists face conflicting temptations: collaboration vs. individual competitive gain.
- ☐ The dilemma shows that individual self-interest often undermines the outcome that is collectively best.
- ☐ Game theory analyzes the strategic interaction (mutual interdependence) inherent in oligopoly.
- ☐ The firms always achieve the cooperative outcome in the long run.
- ☐ Oligopolies are required by law to remain silent about their pricing strategy.

27. Match the game theory term to its definition. /0.5

___ 1. Game Theory

___ 2. Nash Equilibrium

___ 3. Dominant Strategy

___ 4. Prisoner's Dilemma

___ 5. Mutual Interdependence

A. The individual strategy that results in the highest payoff regardless of the rival's decision.

B. The result of players maximizing self-interest, often leading to a suboptimal collective result.

C. The study of strategic interaction where payoffs depend on other players' actions.

D. The characteristic of oligopoly that necessitates the use of game theory analysis.

E. A scenario where cooperation yields larger gains than individual self-interest.

28. The crucial concept that makes analyzing an oligopoly market more complex than analyzing a monopoly or perfect competition is: /0.5

- ◯ The guarantee of zero economic profits in the long run.
- ◯ Homogeneous products.
- ◯ Low barriers to entry.
- ◯ The lack of downward-sloping demand curves.
- ◯ Mutual interdependence, where strategic decisions depend on rivals' choices.

29. Match the economic concept related to pollution with its corresponding definition or effect in the context of production decisions. /0.5

___ 1. Socially Efficient Quantity

___ 2. Social Cost

___ 3. Market Failure

___ 4. Private Cost

___ 5. Negative Externality

A. The sum of a firm's production costs (labor, materials) and the additional harm imposed on society (health, environmental damage).

B. The output level where social costs of production equal the benefits perceived by consumers (demand).

C. Costs directly incurred by the producer in the process of generating output, excluding external impacts.

D. A situation where market mechanisms fail to achieve efficient output because not all costs or benefits are accounted for.

E. A consequence of market exchange that negatively affects a third party who is external to the transaction.

30. Which of the following are the possible ways discussed in this course to address a negative externality like pollution? /0.5

Select all correct answers

- ☐ Pollution Taxes.
- ☐ Marketable Permits.
- ☐ Private parties can work out solutions where the transaction costs of doing so are low, and the relevant property rights are well-defined.
- ☐ Command and Control Regulations.

31. What are shortcomings of Command and Control regulations to protect the environment? /0.5

Select all correct answers

- ☐ they are always cost effective.
- ☐ they offer limited flexibility in where and how pollution will be reduced
- ☐ they provide no incentive to go beyond the limits they set
- ☐ they never have politically motivated loopholes

32. Marketable permits (cap-and-trade systems) are effective mechanisms for reducing pollution because they incorporate which of the following key features? /0.5

Select all correct answers

- ☐ They create a mechanism (a market) where firms with low abatement costs can gain revenue by selling their unused pollution allowances.
- ☐ They ensure that the total quantity of pollution allowed meets the predetermined national pollution standard.
- ☐ They lead to an efficient outcome by guaranteeing that the firms with the highest marginal costs of abatement will reduce pollution the most.
- ☐ They are most appropriate when millions of highly dispersed, small users are responsible for the pollution (e.g., car emissions).
- ☐ They provide firms with flexibility, allowing them to choose whether to abate pollution or to pay for the right to pollute.

33. A municipality is debating a $10 million project to clean a heavily contaminated lake. /0.5

Economists estimate the marginal benefit (MB) of the cleanup is initially very high but declines rapidly as the water quality improves.

They also find that the marginal cost (MC) of filtration increases dramatically as the last few pollutants are removed.

Based on the principle of efficient resource allocation, when should the municipality stop the cleanup project?

- ○ After the first 50% of pollutants are removed, since marginal costs are guaranteed to be lowest during the initial phases.
- ○ When the Marginal Cost of reduction (MC) is exactly equal to the Marginal Benefit of reduction (MB).
- ○ When the total estimated benefits (TB) of the cleanup first equal the total costs (TC) incurred, ensuring the net benefit is zero
- ○ Immediately, because the project costs $10 million, and that money could be spent on life-saving regulations elsewhere.
- ○ Only when the marginal cost of removal approaches infinity, signifying that 100% environmental protection has been achieved.

34. The economic evaluation of U.S. environmental laws suggests that while compliance imposes significant costs on firms, the laws have generated large net benefits for society. /0.5

What are some of the measurable or estimated benefits typically included in this overall calculation?

Select all correct answers

- ☐ The calculation of the value of a statistical human life (VSL) used by agencies like the EPA to quantify health improvements
- ☐ The reduction in social costs resulting from fewer instances of illness and premature death.
- ☐ The increased valuation of assets derived from ecotourism, fishing, or farming sectors reliant on clean resources.
- ☐ The improved quality of life derived from non-marketable activities, such as recreational opportunities.
- ☐ The rising marginal cost curve associated with using increasingly expensive methods to remove the final units of pollution.

35. Which of the following factors contribute to the private sector's tendency to underinvest in research and development (R&D) relative to the needs of society? /0.5

Select all correct answers

- ☐ The difficulty firms face in fully monopolizing the total economic value of their successful R&D efforts.
- ☐ The ability of competing firms to adapt or copy underlying ideas without incurring the original R&D costs.
- ☐ The rivalrous and excludable nature of new technologies once they are successfully implemented.
- ☐ The high fixed costs and inherent uncertainty associated with many R&D projects.
- ☐ The existence of positive externalities, where new innovations provide widespread benefits not captured by the innovating firm.

36. A technology firm has just developed a groundbreaking method for purifying drinking water that is significantly more efficient than existing methods. /0.5

In the absence of any government intervention, competitors could quickly analyze and replicate the core methodology.

Which government policy is primarily designed to transform this technology from a non-excludable concept into a temporary source of monopoly profit for the innovator?

- ◯ Direct government spending to fund the implementation of the new purification method in public water systems.
- ◯ Providing R&D tax credits that reduce the firm's overall tax liability based on its innovation expenditures.
- ◯ Establishing cooperative research ventures between the firm and federal universities
- ◯ Granting intellectual property rights, such as a patent, on the specific purification process.
- ◯ Subsidizing the adoption of the new purification method by consumers to increase market demand.

37. Match the type of government policy used to encourage innovation with a major feature or drawback associated with its implementation. /0.5

___ 1. R&E Tax Credits

___ 2. Cooperative Research Ventures

___ 3. Subsidies (e.g., for education or vaccines)

___ 4. Intellectual Property Rights

___ 5. Direct Government R&D Spending

A. Often criticized for being vulnerable to political decisions regarding which projects receive funding, rather than strictly scientific or economic merit.

B. Can incentivize firms to invest at least a dollar in R&D for every dollar of foregone tax revenue, serving as a cost-effective stimulus.

C. A common response to a positive externality that helps the benefiting party receive a greater share of social benefits, often by lowering the price paid by consumers.

D. Provides the innovator with the exclusive legal ability to make, use, or sell an invention for a set period, thereby allowing temporary monopoly profits.

E. Helps spur product innovation and create new industries by facilitating collaboration between government-funded institutions and the private sector.

38. Consider a local neighborhood organization that relies on voluntary donations to fund enhanced street lighting throughout the community. /0.5

If this street lighting is determined to be nonexcludable and non-rival, the organization will likely struggle to raise sufficient funds due to which of the following phenomena?

- ◯ The free rider problem, where individuals consume the benefit without contributing to the cost
- ◯ The inherent inefficiency that results from providing a club good.
- ◯ The high private rate of return that makes governmental intervention unnecessary.
- ◯ The tragedy of the commons, where the resource is depleted through overuse.
- ◯ The difficulty in defining clear property rights over the service.

39. What are three ways to address the tragedy of the commons? /0.5

Select all correct answers

- ☐ Common Goods
- ☐ Creating Property Rights
- ☐ Free Riders
- ☐ Forced Riders
- ☐ Command and Control regulations requiring licenses, catch limits, and specific seasonal durations.
- ☐ Cultural Norms
- ☐ Private Goods

40. Match the type of good with its unique defining characteristics and associated economic challenge. /0.5

___ 1. Private Good

___ 2. Investment in Education

___ 3. Common Resource

___ 4. Club Good

___ 5. Public Good

A. Nonexcludable and Non-rival, posing the central challenge of the free rider problem for private providers.

B. Excludable and Rival, which allows markets to typically provide these goods efficiently through price mechanisms.

C. Nonexcludable and Rival, leading to the risk of overuse and depletion of the resource, known as the "tragedy of the commons."

D. Excludable and Non-rival, where markets can provide the good, but exclusion may result in some loss of efficiency because the marginal cost of serving an additional consumer is zero.

E. A good characterized by significant positive externalities, though it is fundamentally excludable and can be provided by private firms.